Novel Solid Rocket Propellant Formulations for Low-Cost, Compact Orbital and Suborbital Launch Vehicles

Sachin Shankar • Tesla STEM High School, Redmond, WA • sachinshankar@live.com

ABSTRACT

Access to space has historically been constrained by the enormous cost and scale of launch vehicles. This project investigates novel solid rocket propellant formulations designed to maximize specific impulse (I_{sp}) and thrust-to-weight ratio while minimizing cost and vehicle size, with the goal of enabling smaller, cheaper rockets capable of reaching orbit and delivering payloads. Candidate formulations are evaluated computationally and through controlled small-scale testing, with vehicle integration modeled in OpenRocket and CAD software to validate real-world flight feasibility. The research aims to contribute to the growing body of work on democratized, small-scale access to low-Earth orbit (LEO).

PROBLEM STATEMENT

Current orbital launch vehicles require either expensive liquid propellants with complex plumbing and turbomachinery, or solid propellants that are large, heavy, and prohibitively costly to produce at small scale. There is a significant gap in the market and in open research for compact solid-propellant launch systems capable of reaching orbit or LEO altitudes. The commercial small-satellite industry is growing rapidly — CubeSats, scientific payloads, and technology demonstrators are increasingly common — yet dedicated small-lift launch options remain expensive and scarce. Closing this gap requires propellants that deliver meaningfully higher energy density and lower production cost than conventional formulations, enabling vehicle downsizing without sacrificing mission capability.

HYPOTHESIS

If novel solid propellant compositions can be identified that achieve a meaningfully higher specific impulse (I_{sp}) and energy density than conventional formulations (e.g., standard APCP), then a smaller and cheaper launch vehicle — modeled to reach suborbital or orbital altitudes — can be designed, simulated in OpenRocket, and validated as physically feasible. The cost-per-kg-to-orbit for such a vehicle will be demonstrably lower than that of comparably-sized conventional solid-propellant vehicles.

BACKGROUND & LITERATURE CONTEXT

Solid rocket propellants have been used in launch vehicles since the early Space Age, with ammonium perchlorate composite propellant (APCP) becoming the dominant formula for modern solid boosters. APCP typically delivers I_{sp} values in the range of 250–270 s at sea level, limited by the energy content of its oxidizer-fuel combination. Research into high-energy-density materials (HEDMs) — including aluminum-rich composites, nitrogen-rich compounds, and novel oxidizer combinations — suggests that meaningful performance improvements may be achievable. Simultaneously, advances in small satellite technology have created demand for launch vehicles in the 1–100 kg payload class, a segment where solid propulsion is attractive due to its simplicity and storability. This project sits at the intersection of these two trends.

METHODOLOGY

Phase 1 — Propellant Research & Selection

- Literature review of existing solid propellant chemistries: APCP, HTPB-based composites, nitrate esters, composite metallized fuels, and emerging HEDMs.
- Identification of candidate novel formulations targeting improved I_{sp}, burn stability, and production cost per kilogram.
- Theoretical performance calculations using the rocket propulsion equations: specific impulse, characteristic velocity (c^*), thrust coefficient (C_F), and delivered Δv via the Tsiolkovsky rocket equation.
- Shortlisting of two to four candidate formulations for further simulation and experimental work.

Phase 2 — Simulation & Vehicle Modeling

- Flight trajectory and performance simulation in **OpenRocket** using derived propellant parameters (thrust curve, I_{sp}, propellant mass).
- Vehicle geometry, fin configuration, and structural component design modeled in **CAD software**.
- Iterative optimization of vehicle mass ratio, staging strategy, and nozzle geometry to maximize altitude and payload margin.
- Stability analysis (static margin, Barrowman equations) to ensure aerodynamic feasibility across the flight envelope.

Phase 3 — Experimental Validation

- Small-scale propellant sample preparation and burn-rate testing under controlled, safe conditions.
- Measurement of burn rate, flame temperature (where measurable), and qualitative burn stability.
- Comparison of measured performance metrics against simulated predictions; iterative refinement of formulations.
- Safety and stability assessment of each candidate (sensitivity, aging, compatibility with case materials).

Phase 4 — Analysis & Cost Modeling

- Statistical analysis of burn rate, I_{sp}, and thrust curve data across candidate formulations.
- Cost-per-kg-to-orbit estimation for each optimized vehicle design, benchmarked against conventional approaches.
- Assessment of scalability and manufacturability of top-performing formulations at larger motor sizes.
- Python-based data processing and visualization of all performance curves and cost models.

TOOLS & RESOURCES

Simulation	OpenRocket — flight trajectory, motor performance, stability analysis, altitude prediction
Design	CAD Software — vehicle geometry, fin design, nozzle modeling, structural components
Analysis	Python — data processing, performance curve analysis, cost modeling, visualization
References	NASA Technical Reports Server (NTRS), AIAA Propulsion & Energy literature, Sutton & Biblarz *Rocket Propulsion Elements* (9th ed.), Nakka Experimental Rocketry

EXPECTED OUTCOMES

- Identification of one or more solid propellant formulations with measurably superior I_{sp} and performance-to-cost ratios versus conventional APCP.
- A validated OpenRocket model of a compact launch vehicle capable of reaching suborbital or orbital altitudes on the optimized propellant.
- A cost-per-kg-to-orbit estimate demonstrating the economic viability of the proposed approach relative to existing small-lift launch options.
- A full written research report and experimental dataset suitable for WSSEF and ISEF submission.
- Contribution to open literature on accessible, small-scale solid propellant rocketry.

CURRENT STATUS

This project is currently in progress. Phase 1 literature review is actively underway, with initial focus on APCP baseline characterization and HEDM candidate identification. The OpenRocket simulation environment is configured and baseline vehicle models are under development. CAD modeling of the initial vehicle geometry is ongoing. Propellant candidate shortlisting is expected to be complete before the end of the current academic year.